Lecture Notes

Knowledge Graph Enhanced Educational Notes

1. Data Abstraction

Definition:

The process of hiding irrelevant details from the user. It shows only necessary information to the user and hiding all the irrelevant details from the user.

1.1 Levels of Data Abstraction

Total three levels in database management system: View level, conceptual level and physical level.

- Physical Level: The lowest level of data abstraction. It defines how data are stored, decides location in memory, and whether data will be centralized or distributed.
- Conceptual Level: Defines what data are stored into the database and what are the relationships exist among this data. It decides the structure of entire database.
- View Level: Tells the application about how data should be shown to the user. There are multiple views of the same database.
- Logical Level: Deals with what data are actually stored and the relationship among data stored.

Other Components:

- Data Independence: Users and data should not directly interact with each other.

- Related Concepts

- Database Management System (DBMS): A system where data abstraction will work.
- Irrelevant Details: Details hidden from the user by data abstraction, such as where data will be physically stored or what types of data model will be used.
- Necessary Information: Essential information that users know, not the irrelevant details.
- Complexity: Information to be hidden by data abstraction.
- Essential Details: Provided when hiding complexity.

2. Database Management System (Dbms)

Definition:

A system where data abstraction will work.

3. Database Administrator (Dba)

Definition:

Decides which data should be kept at which particular disk drive. Handles physical level, conceptual level and logical level.

4. Irrelevant Details

Definition:

Details hidden from the user by data abstraction, such as where data will be physically stored or what types of data model will be used.

5. Necessary Information

Definition:
Essential information that users know, not the irrelevant details.

6. User

Definition:
Present on the view level. Users are not directly interact with the physical level.

- Related Concepts
- View Level: Tells the application about how data should be shown to the user. There are multiple views of the same database.
- Physical Level: The lowest level of data abstraction. It defines how data are stored, decides location in memory, and whether data will be centralized or distributed.
- Server: Is a logical level. Fetches the answer of information through the database.

Visual Representation:

Figure 6: Diagram illustrating: Present on the view level. Users are not directly interact with the physical level.

7. Database

Definition:
Directly connected to the physical level. Physical data will be stored at physical level.

8. Programmer/Developer

Definition:
Handle conceptual level and logical level.

9. Database Structure

Definition:
The structure of entire database, including tables and their columns, and relationships between tables, decided in the logical level.

10. Database Queries

Definition:
Different types of queries like Select, Insert, Update, Delete performed in logical level.

11. Security

Definition:
Increases in the view level because authority for editing, changes, and modification is given to specific roles.

12. Server

Definition:
Is a logical level. Fetches the answer of information through the database.

- Related Concepts
- Conceptual Level: Defines what data are stored into the database and what are the relationships exist among this data. It decides the structure of entire database.
- Database: Directly connected to the physical level. Physical data will be stored at physical level.

13. 3 Tier Architecture

Definition:
An architecture seen in the last presentation.

- Related Concepts
- Levels of Data Abstraction: Total three levels in database management system: View level, conceptual level and physical level.

Visual Representation:

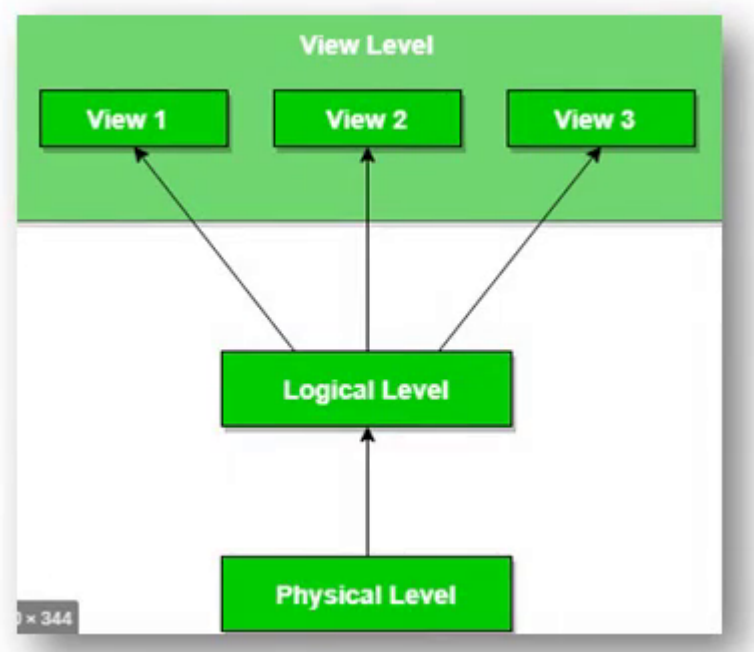

Figure 13: Diagram illustrating: An architecture seen in the last presentation.

14. View Of Data

Definition:

The focus of this presentation.

- Related Concepts

- Multiple Views: Views existing at the presentation level; reasons for needing them are revealed in this presentation.
- Data Abstraction: The process of hiding irrelevant details from the user. It shows only necessary information to the user and hiding all the irrelevant details from the user.

Visual Representation:

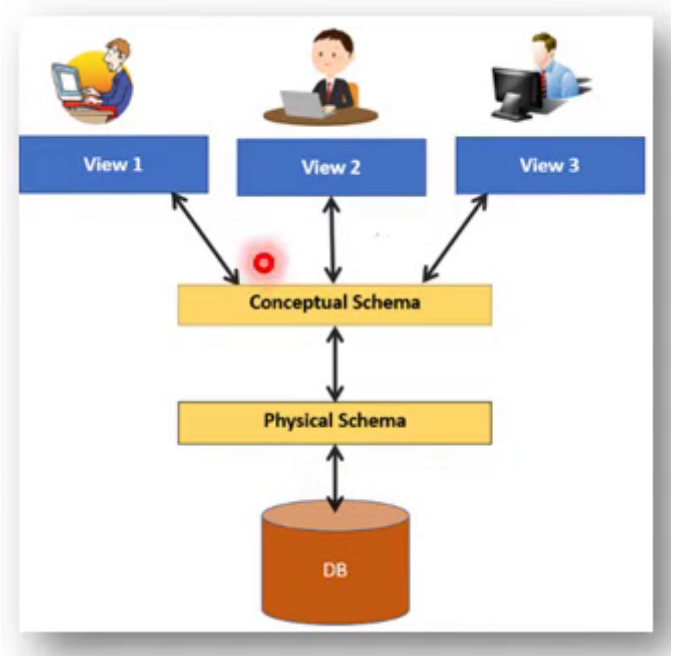

Figure 14: Diagram illustrating: The focus of this presentation.

15. Database Goal

Definition:

To store and retrieve data conveniently and effectively.

- Related Concepts

- Data Storage: Involved for storing and retrieving data.

16. Data Storage

Definition:

Involved for storing and retrieving data.

- Related Concepts

- Storage Data Structures: May involve simple or complex data structures for data storage.

17. Storage Data Structures

Definition:

May involve simple or complex data structures for data storage.

18. Complexity

Definition:

Information to be hidden by data abstraction.

19. Abstract Data

Definition:

A short and sweet answer; desired over complete data.

- Related Concepts

- Complete Data: Not desired; involves encountered complexities.

20. Low Level Data Structures

Definition:

Used at the physical level, especially for multimedia data.

21. What Data Stored

Definition:

Describes the data actually to be stored in the databases.

22. Data Relationships

Definition:

Describes the relationship among the data stored in the database.

23. Physical Data Independence

Definition:

Exists so that if any changes are made in the logical level, it will not affect the physical level.

24. Application Level

Definition:
Where the logical level is, and applications reside.

- Related Concepts
- Schema: The structure of the database.

25. Database Administrator

Definition:
Involved in all levels; writes all the logic, especially at the logical level.

- Related Concepts
- Logical Level: Deals with what data are actually stored and the relationship among data stored.

26. User Access

Definition:
Granting privileges to users, managed at the view level.

27. User-System Interaction

Definition:
Done at the view level through interfaces.

- Related Concepts
- Interfaces: Web, mobile, GUI; used for user interaction with the system.

28. Interfaces

Definition:
Web, mobile, GUI; used for user interaction with the system.

29. Security Of Multiple Views

Definition:
An important aspect to focus on at the view level.